

ZHIYUAN YOU

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Job Objective: Vision (Image & Video) Generation / Editing

Education

- **Ph.D. in Multimedia Lab (MMLab), The Chinese University of Hong Kong** Aug. 2023 -
 - GPA: 3.82/4.0
 - Supervisor: Professor Tianfan Xue & Professor Chao Dong
 - Direction: Generative image processing, multi-modal applications, and low-level vision
- **M.Eng. with Honor, Shanghai Jiao Tong University** Sept. 2020 - Mar. 2023
 - GPA: 3.76/4.0
 - Supervisor: Professor Xinyi Le & Professor Yu Zheng
- **B.Eng. with Honor, Shanghai Jiao Tong University** Sept. 2016 - Jun. 2020
 - GPA: 89.36/100, Ranking: 5/148
 - Supervisor: Professor Xinyi Le
 - Summer Exchange to *Columbia University* in 2018

Experience

- **Seedance Team, ByteDance Seed, Research Intern** Current
 - Direction: Development of the VAE model in video generation model (Seedance series).
 - Direction: Movie & game generation agent in Seedance series.
 - Mentor: Dr. Zhaoyang Huang and Dr. Feng Wang
- **NextCam Team, Adobe, Research Scientist Intern** Jul. 2025 - Nov. 2025
 - Direction: Development of VAE model with large down-sample rate and high dimensional channel.
 - Direction: Image composition guidance based on understanding-generation unified model.
 - Mentor: Dr. Zhoutong Zhang, Dr. Ke Wang, and Dr. He Zhang
- **Shanghai Artificial Intelligence Laboratory, Research Intern** May 2024 - Apr. 2025
 - Mentor: Professor Chao Dong
- **Shenzhen Institutes of Advanced Technology, Research Intern** Apr. 2023 - Jul. 2023
 - Mentor: Professor Chao Dong
- **Horizon Robotics, Research Intern** Nov. 2022 - Mar. 2023
 - Mentor: Dr. Yuelong Yu
- **SenseTime, Research Intern** Dec. 2020 - Nov. 2022
 - Mentor: Mr. Kai Yang

Publications

- M. Yao, **Z. You**, T. Man, M. Wang, T. Xue, “PhotoAgent: Agentic Photo Editing with Exploratory Visual Aesthetic Planning,” in Proc. of International Conference on Machine Learning (**ICML, Oral**), 2026.
- J. Wang, H. Wu, **Z. You**, Y. Song, Y. Wang, Z. Shan, Y. Li, S. Zhang, X. Le, C. Chen, X. Guan, D. Tao, “RouteMoA: Dynamic Routing without Pre-Inference Boosts Efficient Mixture-of-Agents,” in Proc. of Annual Meeting of the Association for Computational Linguistics (**ACL**), 2026.
- X. Cai, **Z. You**, Z. Zhang, T. Xue, “DA-VAE: Plug-in Latent Compression for Diffusion via Detail Alignment,” in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- **Z. You**, K. Wang, H. Zhang, X. Cai, J. Gu, T. Xue, C. Dong, Z. Zhang, “PhotoFramer: Multi-modal Image Composition Instruction,” in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- X. Yin, J. Hu, **Z. You**, K. Yan, Y. Tang, C. Dong, J. Gu, “How Far Have We Gone in Generative Image Restoration? A Study on Its Capability, Limitations and Evaluation Practices,” in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop (**CVPRW**), 2026.
- X. He*, **Z. You***, J. Gong, C. Liu, X. Yue, P. Zhuang, W. Zhang, L. Bai, “RadarQA: Multi-modal Quality Analysis of Weather Radar Forecasts,” in Proc. of Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
- X. Wang, J. Zhuo, **Z. You**, Z. Tan, Y. Yu, S. Wang, X. Le, “ReinAD: Towards Real-world Industrial Anomaly Detection with a Comprehensive Contrastive Dataset,” in Proc. of Annual Conference on Neural Information Processing Systems (**NeurIPS**), Datasets and Benchmarks Track, 2025.
- X. Lin, F. Yu, J. Hu, **Z. You**, W. Shi, J. S. Ren, J. Gu, C. Dong, “Harnessing Diffusion-Yielded Score Priors for Image Restoration,” in ACM **SIGGRAPH Asia**, ACM Transactions on Graphics (**TOG**), 2025.
- J. Gao, R. Liu, Y. Peng, S. Yang, J. Zhang, K. Yang, **Z. You**, “DeQA-Doc: Adapting DeQA-Score to Document Image Quality Assessment,” in Proc. of IEEE/CVF International Conference on Computer Vision Workshop (**ICCVW**), 2025.
- J. Hu*, **Z. You***, J. Gu, K. Zhu, T. Xue, C. Dong, “Revisiting the Generalization Problem of Low-level Vision Models Through the Lens of Image Deraining,” Pattern Recognition (**PR**), 2025.
- J. Hu, J. Gu, S. Yu, F. Yu, Z. Li, **Z. You**, C. Lu, C. Dong, “Interpreting Low-level Vision Models with Causal Effect Maps,” IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2025.
- **Z. You**, X. Cai, J. Gu, T. Xue, C. Dong, “Teaching Large Language Models to Regress Accurate Image Quality Scores using Score Distribution,” in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
- Z. Chen, Y. Wang, X. Cai, **Z. You**, Z. Lu, F. Zhang, S. Guo, T. Xue, “UltraFusion: Ultra High Dynamic Imaging using Exposure Fusion,” in Proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR, Highlight, Best Demo Honorable Mention**), 2025.
- K. Zhu, J. Gu, **Z. You**, Y. Qiao, C. Dong, “An Intelligent Agentic System for Complex Image Restoration Problems,” in Proc. of International Conference on Learning Representations (**ICLR**), 2025.
- J. Zhang*, **Z. You***, J. Wang, X. Le, “SAIL: Sample-Centric In-Context Learning for Document Information Extraction,” in Proc. of Association for the Advancement of Artificial Intelligence (**AAAI, Oral**), 2025.
- **Z. You**, X. Cai, J. Gu, Z. Li, K. Zhu, C. Dong, T. Xue, “Enhancing Descriptive Image Quality Assessment with A Large-scale Multi-modal Dataset,” IEEE Transactions on Image Processing (**TIP**), 2025

- X. Cai, **Z. You**, H. Zhang, W. Liu, J. Gu, T. Xue, “PhoCoLens: Photorealistic and Consistent Reconstruction in Lensless Imaging,” in Proc. of Annual Conference on Neural Information Processing Systems (**NeurIPS, Spotlight**), 2024.
- **Z. You**, Z. Li, J. Gu, Z. Yin, T. Xue, C. Dong, “Depicting Beyond Scores: Advancing Image Quality Assessment through Multi-modal Language Models,” in Proc. of European Conference on Computer Vision (**ECCV**), 2024.
- J. Liu, Y. Zhang, C. Li, **Z. You**, Z. Zhou, C. Yang, Y. Yang, Y. Liu, W. Ouyang, “MaskMA: Towards Zero-Shot Multi-Agent Decision Making with Mask-Based Collaborative Learning,” Transactions on Machine Learning Research (**TMLR**), 2024.
- **Z. You**, K. Yang, W. Luo, X. Lu, L. Cui, X. Le, “Few-Shot Object Counting with Similarity-Aware Feature Enhancement,” in Proc. of IEEE Winter Conference on Applications of Computer Vision (**WACV, Early Accept**), 2023.
- **Z. You**, L. Cui, Y. Shen, K. Yang, X. Lu, Y. Zheng, X. Le, “A Unified Model for Multi-class Anomaly Detection,” in Proc. of Annual Conference on Neural Information Processing Systems (**NeurIPS, Spotlight**), 2022.
- **Z. You**, K. Yang, W. Luo, L. Cui, Y. Zheng, X. Le, “ADTR: Anomaly Detection Transformer with Feature Reconstruction,” in Proc. of International Conference on Neural Information Processing (**ICONIP, Oral**), 2022.
- **Z. You**, J. Li, H. Zhang, B. Yang, X. Le, “An Accurate Star Identification Approach Based on Spectral Graph Matching for Attitude Measurement of Spacecraft,” Complex & Intelligent Systems, 2022.
- L. Chen, **Z. You**, N. Zhang, J. Xi, X. Le, “UTRAD: Anomaly Detection and Localization with U-Transformer,” Neural Networks, 2022.

Services

• Journal Reviewer

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- International Journal of Computer Vision (IJCV)
- IEEE Transactions on Image Processing (TIP)
- IEEE Transactions on Multimedia (TMM)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Systems, Man and Cybernetics: Systems (TSMCS)
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- Pattern Recognition (PR)
- Knowledge Based Systems (KBS)
- Engineering Applications of Artificial Intelligence (EAAI)
- Neural Networks (NN)
- Journal of Intelligent Manufacturing
- Journal of Selected Topics in Signal Processing
- Neurocomputing
- Measurement

• Conference Reviewer

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- International Conference on Computer Vision (ICCV)

- Annual Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- International Conference on Machine Learning (ICML)
- Association for the Advancement of Artificial Intelligence (AAAI)
- ACM International Conference on Multimedia (MM)
- International Conference on Artificial Intelligence and Statistics (AISTATS)
- The British Machine Vision Conference (BMVC)
- IEEE Winter Conference on Applications of Computer Vision (WACV)

Honors and Awards

• CUHK Reaching Out Award	2025
• SJTU Excellent Master Dissertation	2023
• SJTU Outstanding Graduate (Postgraduate)	2023
• National Scholarship	2022
• SJTU Outstanding Graduate (Undergraduate)	2020
• SJTU ABB Scholarship	2017, 2018, 2019
• SJTU Tingya Scholarship	2019
• National Encourage Scholarship	2017

Teaching

• Teaching Assistant, The Chinese University of Hong Kong	Jan. 2025 - May. 2025
– Course: IERG4210, Web Programming and Security.	
• Teaching Assistant, The Chinese University of Hong Kong	Sept. 2024 - Dec. 2024
– Course: IERG4230, Introduction to Internet of Things.	
• Teaching Assistant, The Chinese University of Hong Kong	Sept. 2024 - Dec. 2024
– Course: IERG2051, Signals and Systems.	
• Teaching Assistant, The Chinese University of Hong Kong	Jan. 2024 - May. 2024
– Course: IERG4160, Image and Video Processing.	
• Teaching Assistant, The Chinese University of Hong Kong	Sept. 2023 - Dec. 2023
– Course: ENGG2720, Complex Variables for Engineers.	
• Teaching Assistant, Shanghai Jiao Tong University	Sept. 2020 - Jan. 2021
– Course: ME28009, Dynamic Programming and Optimal Control.	